

RESEARCH METHODOLOGY PAPER

Development and usability of a Hands-On Assessment Tool for Child Health: The HATCH

Soonyoung Shon PhD, RN, FNP, Research Professor¹  |Hyunsook Shin PhD, RN, CPNP-PC, Professor¹  |Yu-nah Lee PhD, RN, Assistant Professor² |Hyunhee Ma RN, MSN, Doctoral Candidate, Researcher¹ |Dahae Rim RN, MSN, Doctoral Candidate, Lecturer¹¹College of Nursing Science, Kyung Hee University, Seoul, Republic of Korea²College of Nursing, Chodang University, Muan, Republic of Korea

Correspondence

Hyunsook Shin, College of Nursing Science, Kyung Hee University, 26 Kyungheedaero, Dongdaemun-gu, Seoul 02447, Republic of Korea.

Email: hsshin@khu.ac.kr

Funding information

Korea International Cooperation Agency, Grant/Award Numbers: #2017-047-01, KOICA #2017-047-01

Abstract

Aim: To develop a comprehensive, hands-on assessment tool for assessing health in children under five in underserved regions.**Design:** Methodological study design and usability testing were used.**Sample:** Eight nurses working at two health posts and 261 children under five living in the migrant villages participated in the study.**Measurement:** The developed tool was evaluated using 10 items of a questionnaire based on the honeycomb model of Morville (2004). Community nurses administered the questionnaire then followed with a focus group interview after completing a child health exam using the developed tool. Data were collected during July 2017.**Results:** The Hands-on Assessment Tool for Child Health is composed of developmental screening, identification of risk factors and clinical signs, growth measurement, diagnostic tests and interpretation of each subdomain, and final impressions. Management strategies include parent education, resource networking, referral to a paediatrician, and follow-up plans. Usability testing revealed high scores on the facets "valuable," "useful," "desirable," and "findable."**Conclusions:** Considering the demand for hands-on tools in underserved regions, the developed tool can provide nurses with resources for competent management of child assessment, interpretation, and nurses' intervention strategies, fortified with clinical judgement processes.

KEYWORDS

assessment tool, child health, community health nursing, health promotion, prevention

SUMMARY STATEMENT

What is already known about this topic?

- Many child health problems are preventable with early detection and intervention. Health problems in children under five affect the growth and development of the whole life.

- Community nurses in underserved regions need comprehensive and hands-on assessment tools with point-of-care to identify child health problems and provide right care.
- Currently developed assessment tools for child health have limitations for use in underserved areas.

What this paper adds?

- The Hands-on Assessment Tool for Child Health (HATCH) helped community nurses to identify the child's health status and nurses to establish a proper care plan.
- The tool provided clinical judgement process of assessment, interpretation, and nurses' intervention strategies.

The implications of this paper:

- The result of the research can contribute to a concrete intervention strategy for child health promotion by assessing health and development in underserved regions.

1 | INTRODUCTION

In recent decades, concerted international efforts have been made to improve children's health status, especially in developing countries, leading to improved statistics in child health, although the World Health Organization (WHO, 2017) reported that health disparities have been widening, depending on location, ethnicity, or socioeconomic status. Among the observed health disparities, malnutrition in children under 5 years in developing countries has been a globally growing concern (Lenters, Wazny, & Bhutta, 2016). Malnutrition causes health problems including stunting, developmental dysfunctions such as deficits in cognition, executive function, and school attainment (M. M. Black et al., 2017), as well as acute, chronic, and infectious diseases (R. E. Black et al., 2013). Because most of these problems are preventable, the WHO (2017) has emphasized the importance of early identification and early intervention.

Kyrgyzstan is one of the developing countries where recent international support for improving children's health status in relation to preventable diseases, specifically, anaemia and stunting, has been provided (SPRING, 2018). However, the National Statistical Committee of the Kyrgyz Republic and United Nations Children's Fund (UNICEF) (2016) reported that about 32.7% of children aged 6–29.9 months in Kyrgyzstan had anaemia and 11.7% were moderately or severely stunted, which indicated a considerable need to improve children's health. Early identification of health risks and early adequate intervention are critical to meet this need.

In contemporary practice, health care providers have been using several tools for identification of health risks. Most of these tools, such as the "30-Second Head-to-toe Tool" (Popovich, 2011) and the "comprehensive health assessment tool" (CHAT) (Chitsabesan et al., 2014), are used in assessing young children's health status or health risk screening. "STRONGkids" is used with in-hospital children to screen for nutritional risk (Hulst, Zwart, Hop, & Joosten, 2010). Current tools have been developed in contexts that are mostly applicable to highly resourced communities such as developed countries or urban areas and are intended for use by specially trained health care providers. Therefore, they are limited in terms of use in underserved

low resourced areas because these tools are complicated to use in remote health clinics with low resources, too fragmented to use in primary care settings, or focused on assessing specific components such as nutrition.

Community nurses, especially those working in underserved regions, need a comprehensive but point-of-care assessment tool including components for risk identification, immediate interpretation, and individual care planning. Recently, technological advances have allowed health care professionals to make immediate as well as adequate clinical judgements, and these include various point-of-care tools. Turner et al. (2016) reported introducing point-of-care tools in primary care settings, especially in their client assessment process, that guide practitioners in decision making and providing adequate yet effective care. Considering that competent clinical judgement is critical to improving children's health in underserved populations, a hands-on assessment tool for children's health equipped with point-of-care tools could enable community nurses working in these environments to take critical roles in improving a target population's health. Therefore, the purpose of this study is to develop a comprehensive and point-of-care child health assessment tool for nurses working in underserved regions.

2 | METHODS

2.1 | Design

This study was conducted in two phases of development and evaluation. A methodological study design and usability testing were used for the development and evaluation phases.

2.2 | Tool development

Item generation

The initial version of the screening tool was developed through item development. A literature review on child health in Kyrgyzstan, using such resources as national statistical reports and UNICEF's Multiple Indicator Cluster Survey (MICS) report, was performed to identify related concepts and items. In addition, site visits and interviews with parents and experts from a community-based child health promotion project site were done by the researchers to identify risk factors of regional child health. The researchers consisted of seven nursing scholars who had experience of conducting studies using focus group interviews (FGIs) and content analysis. The FGI guide was developed based on a literature review and research team discussion. The FGIs were conducted with four groups: one group of 11 parents residing in the migrant villages, one group of child health experts from international organizations and public officers, and two groups of 21 primary health providers working in the primary healthcare units. The health providers (11 nurses and 10 paediatricians) were female, ranging in age from 23–68 years with mean (SD) 48.2 (11.1) years. They had 3–40 years of working experience in primary health care. The

interview questions were factors affecting child health, perceptions of child health promotion, management of high-risk children, quality of child health service in the community health post, and service areas in need. The interviews were translated by interpreters fluent in Kyrgyz, Russian, and Korean and transcribed verbatim by two note-takers. The transcribed statements were analysed using content analysis, and major factors and related items were identified. Using those items, a preliminary tool was developed.

Content validation

The content validity of the preliminary tool was evaluated by seven experts including paediatricians, paediatric nurse practitioners, and experts in child health from Korea and Kyrgyzstan, using 4-point scales of item relevance (Waltz & Bausell, 1981) ranging from 1 (not relevant) to 4 (highly relevant). The content validity indices (CVIs) were calculated for 25 individual items (I-CVIs), the overall scale (S-CVI), and six subscale items. Accuracy, usefulness, and relevance were evaluated for content validity. Modification and deletion were conducted based on expert recommendations, and a final version of the tool, called the "Hands-on Assessment Tool for Child Health" (HATCH) was developed. In addition, nine community nurses performed face validity of the tool. These nurses were female, aged from 23 to 58 years, had graduated from nursing colleges, and had 3–38 years of work experience.

2.2.1 | Usability test

The newly developed HATCH was evaluated using an item response rate and usability testing by community nurses.

Participants

A total of eight community nurses (one of the original group of nurses was on maternal leave) participated in face validity evaluation, and 261 children participated in the evaluation of the HATCH. Nurse participants were responsible for regional children and their family members at a health post of a migrant village, where an official development assistance project was implemented.

Child participants were aged from 6–59 months. They were living in a migrant village, and 123 children were boys, and 138 were girls.

Procedure

We recruited eight nurses working at the project site, a health post that used the developed checklist. To ensure reliability, nurses had three training sessions with the researchers. The training sessions included the presentation of the checklist, whole group question and answer (Q&A), individual and in-pairs exercises, and a final performance test. Before using the tool, a score of 80% for inter-rater agreement among nurses was confirmed.

To apply the developed tool, community health workers (CHWs) recruited 261 children under five registered in two villages from July 2017 (Figure 1). The villages were on the outskirts of Bishkek, the capital of Kyrgyzstan, and established by internal migrants. Therefore,

people residing in those villages were marginalized and disadvantaged (International Federation for Human Rights, 2016). While waiting for a health exam, a receptionist distributed an informational sheet and survey questions including the child's demographics and general characteristics of the family to the parents or guardian.

Nurses had completed 261 well-child check-ups during the whole month of July 2017. After well-child check-ups, the nurses were invited for a survey and a FGI to evaluate the usability of the developed checklist. The survey and subsequent FGI were implemented at a community restaurant. After the survey, using the developed questionnaire based on the honeycomb model, interviews were conducted using 10 FGI guides on the user experience of the checklist. In the subsequent interview, nurses stated their reasons for providing low accessibility scores.

Measures: Usability

Usability means both how easy the programme is to use, as well as how to improve the ease-of-use. Usability is defined as learnability, efficiency, memorability, errors, and satisfaction that users feel when they use a product or service for any purpose (Nielsen, 2012). In this study, the evaluation elements were reconstructed based on the honeycomb model introduced by Morville (2004) to evaluate the usability of the application. In the honeycomb model, user experience was divided into facets of useful, accessible, credible, usable, desirable, findable, and valuable. A 5-point Likert scale using scores from 1 (strongly disagree) to 5 (strongly agree) was used. According to these facets, we developed 10 questionnaire items for the FGI. Useful, credible, usable, and desirable facets had one item, and findable, accessible, and valuable facets had two items. The interview was translated by interpreters fluent in Kyrgyz and Korean and transcribed by two note-takers.

2.3 | Analytic strategy

The newly developed HATCH was evaluated using the item response rate and the honeycomb model. A CVI was used to assess content validity. Frequencies and percentages of participants were used to describe the subdomains and items of the HATCH. To identify the tool's feasibility, complete or incomplete responses to the collected health exam records were estimated. Frequencies and percentages of blank items were calculated. Facets in the usability testing were analysed using means and standard deviations. Additionally, the interview was analysed by each facet item.

2.4 | Ethical considerations

This study was approved by the ethics committees in Kyung Hee University (KHSIRB-17-022(NA)). Parents, guardians, and nurses were requested to give their written informed consent form before participation. Participants were informed through the informational sheet and consent process about any potential risks of participation. All

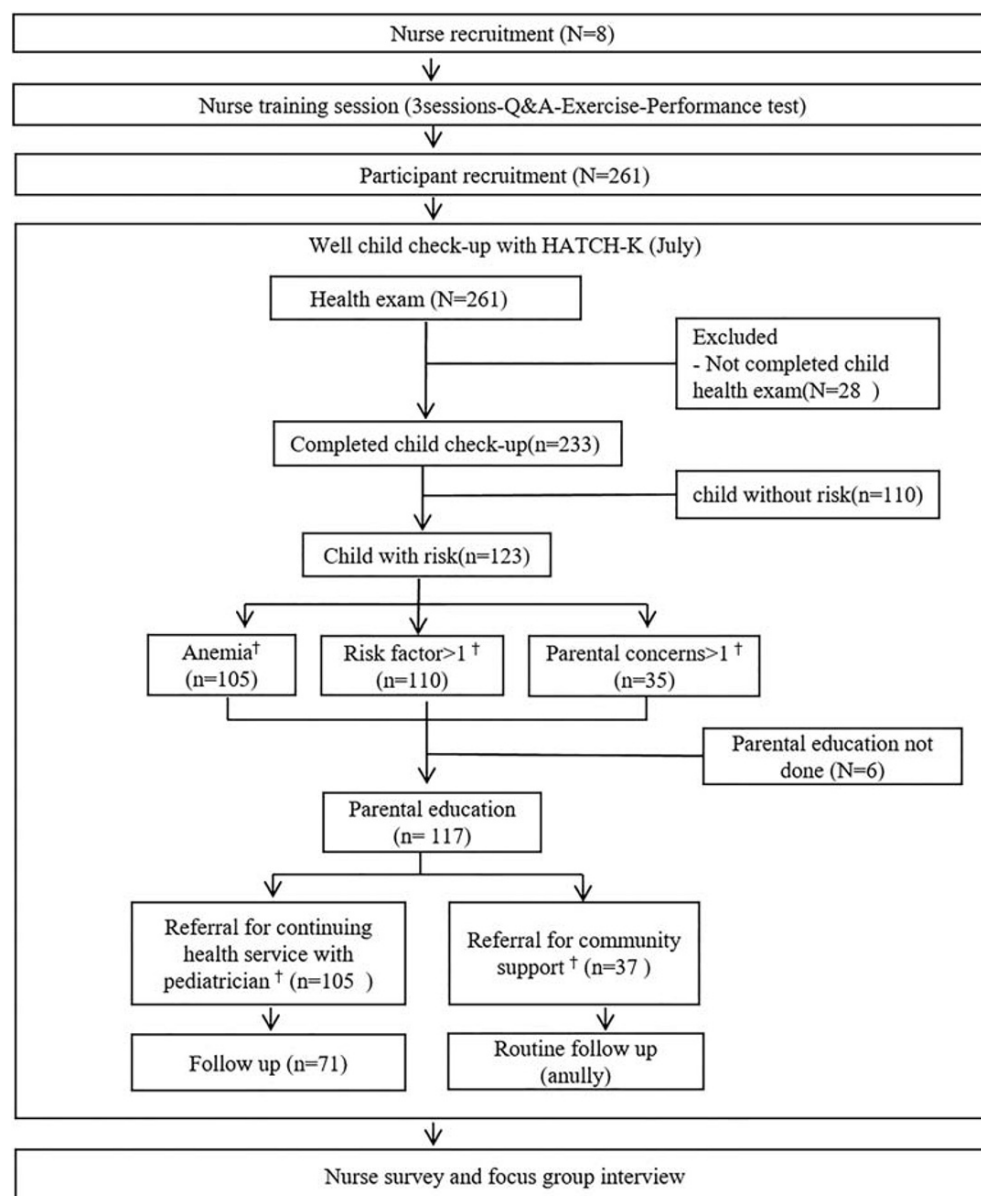


FIGURE 1 The process of research evaluation.[†]Duplicated response item. HATCH, Hands-On Assessment Tool for Child Health; Q&A, question and answer

materials and measures were provided in Russian and Kyrgyz options, and a bilingual medical assistant could answer questions in both languages.

The children who participated in the usability test underwent a health screening. Community nurses managed participants with each management strategy. Participants with anaemia were provided ferrous vouchers according to Kyrgyzstan's treatment guidelines.

3 | RESULTS

3.1 | Development of the Hands-on Assessment Tool for Child Health

The HATCH was initially developed with five subscales and 21 questions, and four different clinical judgement processes and nursing actions corresponded to the interpreted judgments. Assessment items

included a home environment survey, parental concern, anaemia risk factors, growth measurement, physical examination of clinical signs, and diagnostic tests. Every major domain has a judgement point. These included parental concerns, anaemia risk factors, growth measurements, diagnostic tests, and impressions. According to the interpretation results, intervention strategies such as parent education, resource networking, and physician referral were considered. Anthropometric measures, such as weight and height, were subscale items. Weight for age, and height for age and weight were calculated using WHO guidelines. The child health promotion interview subscale consisted of questions about nutrition, elimination, activity, language, and sleeping to identify the caregiver's concerns. The history taking subscale comprised problem symptoms and anaemia risk factors such as premature babies, low birth weight, eating habits, taking nonsteroidal anti-inflammatory drugs (NSAIDs), and taking anthelmintics. The physical examination involved assessing child health and training community nurses through investigation of general appearance, anterior

fontanelle, thrush, mouth fissure, abdominal condition, pallor, body temperature, and heart rate. Haemoglobin and haematocrit were assessed to test for anaemia by using a portable device. Each subscale involved making a judgement according to Kyrgyzstan guidelines.

Seven experts in child health assessed content validity. The CVI of the preliminary HATCH tool was 0.96 for accuracy, 0.97 for usefulness, and 0.96 for relevance. Specifically, the I-CVI, S-CVI, and subscale CVI were 0.96, 1.0, and 0.96, respectively. According to experts' recommendations, items such as a food pyramid and capillary refill time were added for understanding and educational purposes for providers and parents. Other items such as palpation of the anterior fontanelle were eliminated because it was an item available only to children under 24 months of age. Besides, experts recommended that each step needed to have an interpretative phase with an individual subdomain and then a final impression should be made of the accumulated findings from the overall categories. The tool was assessed for face validity by nine community nurses, who reported the item of capillary refill was difficult for nurses to understand, but researchers included the item for training purposes.

The final version of the HATCH consisted of five subdomains and the final nursing impression. Among them, parental concerns about child health, history taking, diagnostic tests, and final nursing impressions contained an item on clinical judgement based on the absence or presence of characteristics in the included items of the subdomain. The final impression was made from accumulated findings from each judgement sector excluding the subdomains of demographics and physical exam. It has corresponding intervention plans depending on the impression. These plans were parent education, resource networking with CHWs, referral to physicians, and follow-up plans.

Scoring methods were made based on the results of four subscales. The first subscale questions asked the child's caregiver about health promotion. Questions concerned nutrition, elimination, activity, language expression and understanding, sleeping, and behaviour. If the caregiver answered "yes" to the question, the nurse provided parents with parental education related to the question area. For example, "Are you worried about your child sleeping?" If a caregiver answered yes, the nurse should educate the caregiver about children's sleeping.

The second subscale questions were about anaemia risk factors. Caregivers could answer either yes or "no." If the caregiver answered yes, the nurse should educate the parent about the questions. Additionally, nurses referred to networking among caregivers and CHWs.

The third subscale items involved anthropometric investigation. Measurement results involved a judgement according to Kyrgyzstan guidelines for stunting. The guidelines were divided into three or four colour zones. Height consisted of the green, sky, and red zones, while weight consisted of the green, sky, yellow, and red zones. The green zone indicated normal range with a z score of 0–2 SD, and the red zone indicated a z score under –2 SD. If the measurement was in the yellow zone, the nurse should educate the caregiver and inform the CHW.

The fourth subscale items concerned an anaemia test. The normal haemoglobin range was based on the child's age. According to the

Kyrgyzstan guidelines, children aged 6 to 24 months with less than 11 mg/dL of haemoglobin were considered to have anaemia, while children aged 25 to 59 months with less than 12 mg/dL were considered to have anaemia. Haemoglobin tests were performed by nurses using a portable device for haemoglobin and haematocrit. If the haemoglobin test showed that the children had anaemia, nurses educated the caregiver, linked the CHW, and referred the child to a primary health doctor.

Finally, the four scoring results determined the overall findings: (a) normal – regular follow up per guidelines, (b) management of high-risk children – nurse education and/or networking with CHW, (c) referral to primary health doctor.

3.2 | Usability of the Hands-on Assessment Tool for Child Health

Table 1 shows the general characteristics of the children and parents who participated in the health check-up. The overall number of recruited children was 261 and 233 screenings: all except 28 incomplete screenings were analysed. Participants in this study were 126 boys (54.1%) and 107 girls (45.9%). The ages of child participants ranged from 6 to 59 months, with an average age of 22.4 months (± 14.93). 34.8% of mothers and 31.3% of fathers were university-graduates, half of the participants owned houses, and half used outdoor toilets.

Table 2 shows the findings of the usability test process and nurses' responses. Among the subdomains and final impression, the most frequently incomplete item was the clinical judgement under the stunting subdomain ($n = 74$), followed by final impression ($n = 66$), risk factors ($n = 55$), and parent concern ($n = 14$). In items on the risk factor subdomain, the item on the presence of low birth weight history remained the highest. Items on the haemoglobin measurement showed complete responses.

A total of 233 children completed the process with the developed HATCH. Of these, 37 children had resource networking, 117 cases had parental education, 105 children were diagnosed with anaemia, and 105 children were referred to paediatricians for their stunting and abnormal diagnostic test results.

Table 3 shows user experiences of the HATCH for eight nurses in Kyrgyzstan. All items averaged above 3.76 ($\pm .68$) in the evaluation. Among these, "valuable" ($4.00 \pm .00$), "useful" ($4.00 \pm .00$), "desirable" ($4.00 \pm .76$), and "findable" ($4.00 \pm .76$) facets scored high. The "accessible" facet scored 3.25 ($\pm .65$), which was the lowest.

A FGI revealed what and why nurses experienced when using the developed tool. Two reversed questions on diet were designated as difficult to mark. Additionally, questions about private matters, such as housing conditions, were identified as difficult to ask. Initially, they had difficulty performing every step of interpretation and planning. In particular, when using community networking for continued child health education/services with CHWs, they did not consider educating parents as part of their role. However, with the training session, they reported that through usability of the tool, they recognized the

TABLE 1 General characteristics of children and parents ($n = 233$)

Item		<i>n</i>	%	Item		<i>n</i>	%
Gender	Male	126	54.1	Owner-occupied housing	Have	125	53.6
	Female	107	45.9		None	57	24.5
Parent occupation	Agriculture	4	2.4	Electricity and utilities in houses	No response	51	21.9
	Business	32	19.2		Have	175	75.1
	Raw labour	30	18.0		None	5	2.1
	Skilled labour	54	32.3	Mother's education level	No response	53	22.7
	Spinning and weaving	13	7.8		Primary school	53	22.7
	Others	34	20.4		College	44	18.9
	No response	66	28.3		University	81	34.8
Expenditure against income	Always in debt.	8	3.4	Father's education level	Post-graduate	2	0.9
	Often in debt.	62	26.6		No response	53	22.7
	Adequate	101	43.3		Primary school	62	26.6
	Remained	1	3.0		College	33	14.2
	No response	55	23.6		University	73	31.3
Expenditure against food expenses	Less than half	9	3.9	Toilet location	Post-graduate	1	0.4
	Half	49	21.0		No response	64	27.5
	More than half	51	21.9		Inside	57	24.5
	Almost	28	12.0		Outside	119	51.1
	Do not know	38	16.3		No response	57	24.5
	No response	58	24.9				

TABLE 2 Findings of the usability test process and nurses' responses (feasibility and item response rate) ($n = 233$)

		Responded item			
		Health risk (identified)			
Subdomain	Item	Presence	Absence	Total	Blank item
Parent concern	Clinical judgement	35	184	219	14
Risk factor	Clinical judgement	110	68	178	55
	Low birth weight	10	160	170	63
	Preterm	57	121	178	55
	Balanced diet	85	142	227	6
	Eating habit	105	110	215	18
	Smoking	79	139	218	15
	Pica	41	177	218	15
	Hygiene	37	192	229	4
Stunting	Z score	5	154	88	74
	Weight	N/A	N/A	228	5
	Height	N/A	N/A	230	3
Diagnostic test	Clinical judgement	105	128	233	0
	Haemoglobin	N/A	N/A	233	0
	Haematocrit	N/A	N/A	226	7
Final nursing impression/plans	Overall nursing impression	123	44	167	66
	A. Parent education	117	-	-	-
	B. Resource networking with CHWs	37	-	-	-
	C. Referral to paediatricians	Refer105 Fu71	-	-	-

Abbreviation: CHWs, community health workers.

TABLE 3 Usability test of HATCH (*n* = 8)

Facets of usability	Content	Mean $\pm$ SD	Min	Max
Useful	Frequency of HATCH	4.00 $\pm$ 0.00	4	4
Accessible	Readability of items	3.00 $\pm$ 1.20	1	4
	Easy operation of the main menu	3.50 $\pm$ 0.53	3	4
	Mean	3.25 $\pm$ 0.65		
Credible	Reliable health assessment of child	3.75 $\pm$ 0.46	3	4
Usable	Easy to fulfil	3.38 $\pm$ 0.52	3	4
Desirable	Combination of tables and text	4.00 $\pm$ 0.76	3	5
Findable	Location of menu	4.00 $\pm$ 0.76	3	5
	Search function	4.00 $\pm$ 0.54	3	5
	Mean	4.00 $\pm$ 0.60		
Valuable	Recommend to other nurses	4.13 $\pm$ 0.35	4	5
	Recommend to other regions	3.88 $\pm$ 0.35	3	4
	Mean	4.00 $\pm$ 0.00		

Abbreviation: HATCH, Hands-On Assessment Tool for Child Health.

importance and effectiveness of parental education provided by community nurses.

4 | DISCUSSION

The strength of the newly developed HATCH is that it contains steps of clinical judgement in each subdomain as well as comprehensive components of child health screening and management strategies. Nurses working in underserved regions, such as migrant villages, need to have comprehensive but efficient tools to provide child care. Current child assessment tools such as "CHAT," 30-Second Head-to-Toe Tool, and STRONGkids are limited in meeting the demand for primary health care in regions with low resources (Chitsabesan et al., 2014; Hulst et al., 2010; Popovich, 2011). CHAT (Chitsabesan et al., 2014) was developed for adolescent offenders in England and contained 4 subdomains of screening (ie, physical health, mental health, substance misuse, and neuro-disability). In CHAT, the identified needs are summarized at the end of each section, which resulted in care planning of referral to specialists. However, it takes a long time (about 85 min) and needs 4 types of trained professionals for completion. On the other hand, the 30-Second Head-to-Toe Tool (Popovich, 2011) was developed for paediatric nurses to quickly assess a child's condition and surrounding environment. STRONGkids (Hulst et al., 2010) is used for nutritional risk screening with in-hospital children and has risk management plans, such as referral to a doctor or dietician. Therefore, neither the 30-Second Head-to-Toe Tool nor STRONGkids are comprehensive enough to use in low-resource areas. The HATCH is a comprehensive screening tool that includes the process of developmental screening, identification of risk factors and clinical signs, growth measurement, diagnostic test, and impression. Furthermore, it has management strategies such as parent education, resource networking, referral to a paediatrician, and follow up plans depending on the step of the judgement process.

The HATCH comprises 5 subdomains in its measurement items: parents' concern, risk factors, growth measurement, physical examination, and diagnostic test. The first subdomain, Parents' concerns child development, consists of eating, elimination, activity, language, sleeping, and behaviour. Parents' concern was a significant indicator of children's health status, reflecting the sensitivity and responsibility of parents for their children (<https://doi.org/10.1016/S0140-6736, 2007>). Schaffer, Goodhue, Stennes, and Lanigan (2012) suggested that enhancing responsive parenting was an effective nursing strategy. The second and third subdomains were identification of risk factors and clinical signs. Included risk factors were birth history, balanced diet, patterns of tea consumption, intake of flour and salt with zinc and iodine, family smoking, presence of pica, and hygiene. These are intended to assess the risk of undernutrition, parasite infection, and micronutrient deficiency inducing anaemia and stunting. Iodine, zinc, vitamin B12, and iron are associated with neurodevelopment during early stages (John, Black, & Nelson, 2017), and iodine deficiency especially could lead to congenital hypothyroidism and irreversible mental retardation, making it the most common preventable cause of mental retardation (<https://doi.org/10.1016/S0140-6736, 2007>). Poor hygiene and parasite infection are the causes of malnutrition and anaemia, and pica is a clinical sign of parasite infection. The fourth subdomain of growth measurement was to assess whether growth was optimal or stunted. In a meta-analysis in low- and middle-income countries, stunting in young children was negatively associated with on-track development, especially physical development, literacy/numeracy development, and learning (Miller, Murray, Thomson, & Arbour, 2016). Another study showed that underweight and stunting were associated with apathy, less positive affect, lower levels of play, and more insecure attachment than in non-growth-retarded children (<https://doi.org/10.1016/S0140-6736, 2007>). The fifth subdomain was a diagnostic test for anaemia using a portable device (CERA-CHEK Hb Plus, Greencross Medisys, Korea) as a point-of-care. The device could investigate haemoglobin and haematocrit in seconds

by a finger prick method. Using a point-of-care device can enable nurses to provide parents with an effective management plan.

The HATCH had horizontal and vertical directions for making a judgment. Horizontally, each subdomain contained steps of the clinical judgement process leading to nurses' intervention strategies. Vertically, it was organised to perform a comprehensive child health assessment. Compared with other health assessment tools (Chitsabesan et al., 2014; Popovich, 2011) that proceed linearly, the HATCH is multifaceted, so it can be used for training purposes to enhance nurses' clinical judgement. The newly developed tool aims to provide opportunities for parental education on child health using the child health exam as well as to screen children's nutritional and general health status. When it was applied to a child population in a community health post, nurses reported that the general format and items of the checklist were familiar in their practice, although they preferred this comprehensive type of checklist that systematically assesses a child's health. Each subsection requires performing a semi-final interpretation called clinical judgement. Nurses performing health exams using the tool frequently had difficulty completing the judgement process in the very beginning, but with more practice, the completion rate of the checklist increased.

Findings of the study showed that the HATCH identified 123 children as at-risk. These were 35 cases reporting parental concern, 110 cases presenting anaemia risk factors, 5 cases of stunting, and 105 abnormal findings from diagnostic tests. Compared with previous studies presenting the evaluating efficiency of screening tools (Raja et al., 2008), the findings of those reporting cases using the HATCH show the effectiveness of the tool.

Incomplete responses in items on clinical judgement reflect nurses' limited capacity and their need for more training on how to decide on their assessment findings. The incomplete items identified which item were difficult for nurses, and this could help in planning nurses' training. Furthermore, when evaluating the tool using item response, the most frequently incomplete item was clinical judgement whereas the most complete one was measuring the level of haemoglobin. This occurred because nurses in the study experienced challenges in making clinical judgments in a timely manner. Additionally, in usability testing, the findings demonstrated that the HATCH was convenient to use, credible to measure, and valuable enough to recommend to other nurses. Nurses participating in the evaluation stated that the tool needed to be updated to improve accessibility and usability through revising the questions and developing a more convenient way of approaching the judgement process.

4.1 | Strengths and limitations

This study is important because it can contribute to the reduction of health disparities through an effective assessment tool for use at point-of-care in underserved regions. This multifaceted (vertical and horizontal) tool can provide community nurses with a hands-on as well as comprehensive tool for child health assessment. The other strength of the study is in its actual evaluation among a population in an

underserved region. However, this study has several limitations. First, a small number of community nurses participated in the evaluation process. Second, the tool needs to be evaluated its diagnostic accuracy of haemoglobin and haematocrit. In addition, the tool was developed to target a specific region, which may limit its further application to other regions. Hence, the findings of the study need to be interpreted carefully considering these limitations.

5 | CONCLUSION

Nurses working in a community environment such as migrant villages in this study, need to have comprehensive as well as effective tools to provide proper child care. The HATCH may help nurses working in underserved regions to undertake the comprehensive clinical judgement process of child assessment, interpretation, and nurses' intervention strategies for identifying the risk factors of child health and nutrition.

ACKNOWLEDGEMENTS

We thank all children and nurses participating in this study.

CONFLICT OF INTEREST

The authors have no conflicts of interest to report. All listed authors are in agreement with the content of the manuscript.

AUTHOR CONTRIBUTIONS

Shin H. contributed to conception and design, analysed the data and interpretation, drafted the manuscript, critically revised the manuscript, and gave final approval. Shon S. contributed to conception, design, and acquisition, analysed the data and interpretation, drafted the manuscript, critically revised the manuscript, and gave final approval. Lee Y. contributed to conception and design, drafted the manuscript, and critically revised the manuscript. Ma H. contributed to conception and design, analysed the data and interpretation, drafted the manuscript, and critically revised the manuscript. Rim D. contributed to design and acquisition, analysed the data and interpretation, drafted the manuscript, critically revised the manuscript and gave final approval.

FUNDING INFORMATION

This study was supported by Korea International Cooperation Agency (KOICA #2017-047-01).

ORCID

Soonyoung Shon  <https://orcid.org/0000-0003-0571-2181>

Hyunsook Shin  <https://orcid.org/0000-0002-2795-5582>

REFERENCES

- Black, M. M., Walker, S. P., Fernald, L. C. H., Andersen, C. T., DiGirolamo, A. M., Lu, C., ... Grantham-McGregor, S. (2017). Early childhood development coming of age: science through the life course. *The Lancet*, 389(10064), 77–90. [https://doi.org/10.1016/S0140-6736\(16\)31389-7](https://doi.org/10.1016/S0140-6736(16)31389-7)

- Black, R. E., Victora, C. G., Walker, S. P., Bhutta, Z. A., Christian, P., De Onis, M., ... Uauy, R. (2013). Maternal and child undernutrition and overweight in low-income and middle-income countries. *The Lancet*, 382(9890), 427–451. [https://doi.org/10.1016/S0140-6736\(13\)60937-X](https://doi.org/10.1016/S0140-6736(13)60937-X)
- Chitsabesan, P., Lennox, C., Theodosiou, L., Law, H., Bailey, S., & Shaw, J. (2014). The development of the comprehensive health assessment tool for young offenders within the secure estate. *The Journal of Forensic Psychiatry & Psychology*, 25(1), 1–25. <https://doi.org/10.1080/14789949.2014.882387>
- Walker, S. P., Wachs, T. D., Meeks Gardner, J., Lozoff, B., Wasserman, G. A., Pollitt, E., & Carter, J. A. (2007). Child development: risk factors for adverse outcomes in developing countries. *The Lancet*, 145–157. [https://doi.org/10.1016/S0140-6736\(07\)60076-2](https://doi.org/10.1016/S0140-6736(07)60076-2)
- Hulst, J. M., Zwart, H., Hop, W. C., & Joosten, K. F. M. (2010). Dutch national survey to test the STRONGkids nutritional risk screening tool in hospitalized children. *Clinical Nutrition*, 29(1), 106–111. <https://doi.org/10.1016/j.clnu.2009.07.006>
- International Federation for Human Rights, FIDH. (2016). Kyrgyzstan women and children from Kyrgyzstan affected by migration. Retrieved from https://www.fidh.org/IMG/pdf/rapport_kyrgyzstan_uk-2-web2.pdf
- John, C. C., Black, M. M., & Nelson, C. A. (2017). Neurodevelopment: the impact of nutrition and inflammation during early to middle childhood in low-resource settings. *Pediatrics*, 139(Supplement 1), S59–S71. <https://doi.org/10.1542/peds.2016-2828H>
- Lenters, L., Wazny, K., & Bhutta, Z. A. (2016). Management of severe and moderate acute malnutrition in children. In *Reproductive, Maternal, Newborn, and Child Health*, Third Edition (Volume 2). Washington (DC): The International Bank for Reconstruction and Development / The World Bank.
- Miller, A. C., Murray, M. B., Thomson, D. R., & Arbour, M. C. (2016). How consistent are associations between stunting and child development? Evidence from a meta-analysis of associations between stunting and multidimensional child development in fifteen low- and middle-income countries. *Public Health Nutrition*, 19(8), 1339–1347. <https://doi.org/10.1017/S136898001500227X>
- Morville, P. (2004). User experience honeycomb. *Web Log Post* Retrieved from <http://Semanticstudios.Com/Publications/Semantics/000029.Php>, 21, 9–12.
- National Statistical Committee of the Kyrgyz Republic, & UNICEF (2016). *Kyrgyzstan Multiple Indicator Cluster Survey 2014, Final Report*. Kyrgyzstan: Bishkek.
- Nielsen, J. (2012). How many test users in a usability study. *Nielsen Norman Group*, 4(06). Retrieved from <https://www.nngroup.com/articles/how-many-test-users/>
- Popovich, D. (2011). 30-Second Head-to-Toe Tool in pediatric nursing: cultivating safety in handoff communication. *Continuing Nursing Education*, 37(2), 55–60.
- Raja, R., Gibson, S., Turner, A., Winderlich, J., Porter, J., Cant, R., & Aroni, R. (2008). Nurses' views and practices regarding use of validated nutrition screening tools. *Australian Journal of Advanced Nursing*, 26(1), 26–33.
- Schaffer, M. A., Goodhue, A., Stennes, K., & Lanigan, C. (2012). Evaluation of a public health nurse visiting program for pregnant and parenting teens. *Public Health Nursing*, 29(3), 218–231. <https://doi.org/10.1111/j.1525-1446.2011.01005.x>
- SPRING (2018). *Endline nutrition survey in the Kyrgyz Republic: analytical report*. VA: Arlington.
- Turner, P. J., Van den Bruel, A., Jones, C. H., Plüddemann, A., Heneghan, C., Thompson, M. J., ... Howick, J. (2016). Point-of-care testing in UK primary care: a survey to establish clinical needs. *Family Practice*, 33(4), 388–394. <https://doi.org/10.1093/fampra/cmw018>
- Waltz, C. F., & Bausell, B. R. (1981). *Nursing research: design statistics and computer analysis*. Davis FA.
- World Health Organization (2017). *Nutritional anaemias: tools for effective prevention*. World Health Organization.

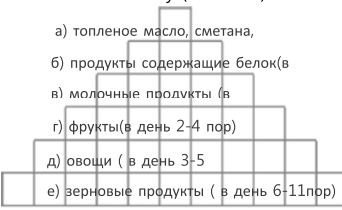
How to cite this article: Shon S, Shin H, Lee Y-n, Ma H, Rim D. Development and usability of a Hands-On Assessment Tool for Child Health: The HATCH. *Int J Nurs Pract*. 2020;26:e12826. <https://doi.org/10.1111/ijn.12826>

APPENDIX 1 HATCH-K A.

HATCH-K

ID			Village	☐ A ☐ B ☐ C ☐ D ☐ 1 ☐ 2 ☐ 3 ☐ 4 ☐ 5	
Mother's Name					
Child's Name					
Child's Gender	☐ Male ☐ Female		Child's birthday	(yyyy/mm/dd)	
Screening No	☐ 1 ☐ 2 ☐ 3 ☐ 4 ☐ 5		Screening date	(yyyy/mm/dd)	
Screening place	☐ home visit ☐ poly clinic		Examiner's name		

Parents concern	Nutrition	Excretion	Activity	Language	Sleeping	Behavior	concern ☐ Yes → Parents education
(if yes, check the box)	☐	☐	☐	☐	☐	☐	

Risk factor of Anaemia	1. Was the child born within 37wks? ☐ Yes ☐ No 2) Birth weight ☐ under 2.5kg ☐ more than 2.5kg (actual: _____ kg) 2-1. Dietary Intake: What your child has eaten in the last day (24 hours)? 		2-2. Dietary habit 1) Do you use tea to prepare your child's milk? ☐ Yes ☐ No 2) Do you use flour that contains no zinc or iodine? ☐ Yes ☐ No 3) Do you use salt that contains no zinc or iodine? ☐ Yes ☐ No	
	3. Does anyone in your family smoke? ☐ Yes ☐ No 4-1. Does your child eat uneatable things other than fruits and meals? (e.g. soil, tree, etc) ☐ Yes ☐ No 4-2. Does your child have a habit of sucking furniture, toys, or paint? ☐ Yes ☐ No 5. Hand Washing of Parents: Hygiene -After using toilet ☐ Yes ☐ No -Before cooking. ☐ Yes ☐ No - When came back after going out ☐ Yes ☐ No		Risk Factor ☐ Yes → Parent education and refer to CHW ☐ No Yes: More than 1 risk factor	

Measureme nt of child growth	Height	cm	Weight	kg
	(zone:)		(zone:)	

Physical Assessment	General appearance	Ill Looking ☐ Yes ☐ No	Peripheral edema ☐ Yes ☐ No
	Body Temp. °C	HR beat/min	Murmur ☐ Yes ☐ No
	Abdominal tenderness ☐ Yes ☐ No		
	Mouth	Thrush ☐ Yes ☐ No	Mouth Fissure ☐ Yes ☐ No
	conjunctiva pallor	☐ Normal ☐ Abnormal	Capillary Refill Time ☐ Normal ☐ Abnormal

Screening of Anaemia	Haemoglobin g/dL	Haematocrit %

Summarize	☐ Normal (next visit:)	☐ Risk manage group ☐ Parent education / ☐ refer to CHW	☐ Refer group (refer to doctor)
-----------	--	---	--